

Domain-Specific RAG System for Healthcare

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Github Repo Link: github.com/srpra/291A-RAG-Project/tree/main

1. Introduction

Large Language Models (LLMs) are increasingly capable of understanding and generating human-like text, yet they often lack factual grounding, especially in specialized domains such as healthcare. Retrieval-Augmented Generation (RAG) mitigates this limitation by coupling a retriever with a generator, ensuring that outputs reference up-to-date and verified documents rather than relying solely on internal parameters. Healthcare is a high-stakes domain where accuracy, timeliness, and transparency are critical. Outdated or hallucinated information can directly impact patient outcomes. Integrating retrieval with generation enables systems that deliver medically sound, evidence-based answers grounded in trusted sources such as clinical guidelines, peer-reviewed studies, and WHO publications.

2. Domain Selection and Justification

2.1 Why Healthcare?

Healthcare is one of the most information-intensive fields, spanning research papers, clinical notes, and policy reports. Clinicians and researchers continuously seek verified data for diagnosis, treatment, and decision support. A healthcare-specific RAG system bridges these dispersed resources, providing contextualized access to knowledge and ensuring safety in downstream generation.

2.2 Retrieval Challenges

- Terminological variability: e.g., myocardial infarction vs. heart attack.
- Context dependence: relevance varies with patient age or region.
- Have to deal with structured as well as unstructured data
- High precision requirement: medical errors are unacceptable

2.3 Benefits of RAG in Healthcare

RAG ensures factual reliability by retrieving from verified sources, producing concise, source-linked explanations. It supports explainable AI and improves trust in medical decision-support tools.

2.4 Key Stakeholders

Primary: Medical students (learning), researchers (literature review), and administrators (auditing).

Secondary: Doctors and clinicians verifying treatments and diagnoses.

3. Dataset Construction and Request Set Design

A corpus of 119 documents was compiled, consisting of 89 PDF research papers and 30 TXT articles and reports. Documents were sourced from PubMed Central, WHO, and other open-access journals. 11 queries were created to reflect authentic healthcare information needs, ranging from factual to analytical questions. They test semantic retrieval beyond keyword matching and require cross-document reasoning.

4. Baseline System and Evaluation

The retrieval backend used FAISS (Facebook AI Similarity Search) with SentenceTransformers 'all-MiniLM-L6-v2' embeddings. Each document was encoded as a dense vector, and FAISS performed

cosine-similarity search to return the top-k matches per query, where the output is evaluated across multiple values of k. For each query, five manually ranked relevant documents formed a golden set. Evaluation used Precision@k, Recall@k, F1@k, MRR, latency and nDCG@k to quantify retrieval effectiveness. These metrics were chosen to comprehensively evaluate retrieval performance, measuring accuracy (Precision), completeness (Recall), balance (F1), ranking quality (MRR), and relevance ordering (nDCG) and time taken for retrieval (latency).

Example query: Based on the collective perspective of recent literature, what are the major biological, technological, and sociopolitical barriers that still hinder full malaria eradication despite modern advances? (for this example, k=7 of all values is chosen)

Retrieved Documents:

1. An Overview of Malaria Transmission Mechanisms, Control,.txt
2. Malaria - Past, Present and Future.pdf
3. WHO-guidelines-for-malaria.pdf
4. Malaria Vaccines - Recent Advances and New Horizons.pdf
5. Malaria Pathogenesis Danny A.txt
6. Malaria - An Overview.pdf
7. History of the discovery of the malaria parasites and their vectors.pdf

Golden Set:

1. WHO-guidelines-for-malaria.pdf
2. Malaria - Past, Present and Future.pdf
3. An update on Malaria.txt
4. Malaria - An Overview.pdf
5. Malaria Vaccines - Recent Advances and New Horizons.pdf

Score:

Metric (k=7)	Value
Precision@7	0.57
Recall@7	0.8
f1@7	0.66
nDCG@7	0.81
MRR	0.5
Latency	0.0404

5. Conclusion

This phase established a comprehensive healthcare dataset, a realistic query set, and a quantitative evaluator. Results demonstrate that RAG retrieval can reliably surface accurate medical information from heterogeneous sources. With improved indexing and generation, the system can serve as a clinically grounded AI assistant for healthcare professionals and learners.

Appendix: Supplementary Materials

A. Complete list of requests

1. Which groups are considered at highest risk of severe influenza complications, and what medical or preventive interventions are recommended for them?,

2. What are the major risk factors and early symptoms of myocardial infarction, and how does public awareness impact treatment outcomes?,
3. How can early diagnosis, evolving clinical definitions, and global prevention strategies improve sepsis outcomes and reduce mortality worldwide?,
4. Based on the collective perspective of recent literature, what are the major biological, technological, and sociopolitical barriers that still hinder full malaria eradication despite modern advances?,
5. What evidence exists across sources that mass vaccination programs have transformed global mortality and morbidity patterns, and what indicators suggest this progress is at risk?,
6. How do logistical, social, and structural barriers affect vaccine delivery in rural or underserved regions, and what community-based interventions have shown promise in mitigating these obstacles?,
7. Which emerging therapies, including gene therapy and molecular targeted approaches, show promise in revolutionizing overall cancer treatment paradigms?,
8. Which innovations in smart healthcare systems are improving disease diagnosis and clinical decision-making?,
9. What key social and environmental factors have been identified as influencing mental health outcomes, and how do recent studies suggest addressing them?,
10. What role do digital technologies and mobile platforms play in improving diabetes management and patient engagement?,
11. What is insulin? How does it regulate blood glucose, and how has insulin therapy evolved?

B. Tentative Timeline:

1. **Week 5, 6:** Test domain-specific embeddings, compare with baseline embeddings and experiment with different FAISS indexing strategies and similarity metrics.
2. **Week 7, 8:** Experiment with query enhancements, hybrid embeddings, and re-ranking strategies to improve retrieval accuracy.
3. **Week 9:** Integrate best-performing embeddings and retrieval strategies into the FAISS pipeline, optimize parameters, and run full evaluation on all queries.
4. **Week 10:** Analyze metric-wise performance (Precision@k, Recall@k, MRR, nDCG), finalize system configuration, and complete documentation/GitHub submission.